Standard Operating Procedure for Resolving BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding on Cisco PE Device MAR-PE-13 at Marathahalli Core

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_1094

Description

BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on Cisco PE devices in the Marathahalli Core network can occur due to various reasons including but not limited to, misconfigured routing protocols, neighbor issues, and suboptimal routing policies. This SOP outlines the steps to diagnose and resolve these issues on the MAR-PE-13 device. The device is responsible for maintaining routing information and providing connectivity between different networks. The BGP route churn can be caused by frequent updates to the routing table, OSPF adjacency loss can be due to issues with the OSPF neighbor relationships, and ISIS LSP flooding can be caused by misconfigured ISIS metrics or other network issues. The Cisco PE device uses the OSPF, ISIS, and BGP routing protocols to maintain routing information and provide connectivity between different networks. The device also uses the Cisco IOS software to manage the routing protocols and provide features such as route filtering and redistribution. The MAR-PE-13 device is connected to multiple networks and provides connectivity between them. The device has multiple interfaces, including Ethernet and POS interfaces, which are used to connect to different networks. The device also has multiple routing protocols enabled, including OSPF, ISIS, and BGP.

Pre-requisite Checks

- Verify that the MAR-PE-13 device is physically accessible at the Marathahalli Core site.

- Check the redundant link status to ensure that there are no issues with the backup links.

- Verify that the device is running the correct software version and that there are no software bugs that could be causing the issue.

Potential Root Causes

Cause	Probability
Misconfigured routing protocols	High
Neighbor issues	Medium
Suboptimal routing policies	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 14, main routing table version is 14 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State PrepQ Done Rcv-Only neighbor1 4 65010 0 0 0 0 0 00:00:00 Active 0 0 No neighbor2 4 65011 0 0 0 0 0 00:00:00 Active 0 0 No, Description: This command displays a summary of the BGP neighbor status, including the version number, AS number, message received and sent, table version, input and output queues, and up/down time.

Command: show ip ospf interface, Expected Output: OSPF Process ID 1, Router ID 10.10.10.1, Network Type BROADCAST Interface IP Address Process ID Interface Cost State Nbrs F/C Loopback0 10.10.10.1 1 0 1 DR 0 0 Ethernet0 10.10.10.2 1 0 1 DR 1 0 POS0 10.10.10.3 1 0 1 DR 2 0, Description: This command displays a summary of the OSPF interface status, including the interface name, IP address, process ID, interface cost, state, number of neighbors, and whether the interface is in the forward or clear state.
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Resolution Steps

Step 1: Step: Check the BGP neighbor status, Command: show ip bgp summary, Description: Verify that the BGP neighbors are in the active state and that there are no errors or issues with the neighbor relationships.

Step 2: Step: Check the OSPF interface status, Command: show ip ospf interface, Description: Verify that the OSPF interfaces are in the DR state and that there are no errors or issues with the interfaces.

Step 3: Step: Configure the routing protocols, Command: router ospf 1 network 10.10.10.0 0.0.0.255 area 0 router bgp 65010 neighbor 10.10.10.2 remote-as 65011, Description: Configure the OSPF and BGP routing protocols on the MAR-PE-13 device, including setting the process ID, network, and neighbor IP addresses.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 15, main routing table version is 15 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State PrepQ Done Rcv-Only neighbor1 4 65010 0 0 0 0 0 00:00:01 Active 0 0 No neighbor2 4 65011 0 0 0 0 0 00:00:01 Active 0 0 No, Description: Verify that the BGP neighbor status has been resolved and that there are no errors or issues with the neighbor relationships.

Command: show ip ospf interface, Expected Output: OSPF Process ID 1, Router ID 10.10.10.1, Network Type BROADCAST Interface IP Address Process ID Interface Cost State Nbrs F/C Loopback0 10.10.10.1 1 0 1 DR 0 0 Ethernet0 10.10.10.2 1 0 1 DR 1 0 POS0 10.10.10.3 1 0 1 DR 2 0, Description: Verify that the OSPF interface status has been resolved and that there are no errors or issues with the interfaces.
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Roll-back Procedures

- Step: Undo the routing protocol configuration, Command: no router ospf 1 no bgp 65010, Description: Undo the routing protocol configuration on the MAR-PE-13 device, including removing the OSPF and BGP processes.